

The sharp multiplier threshold on $N_{3,2}$ is determined by arXiv:1508.01687

Literature-status note for `fa_banach_001`

17 August 2026

Status. The sharpness conjecture in arXiv:1210.8090 is answered affirmatively by the general two-step lower bound of Martini–Müller, arXiv:1508.01687, Theorem 3. The later paper also explicitly lists the $N_{3,2}$ theorem among the earlier results that this lower bound makes sharp.

Let L be the standard homogeneous sub-Laplacian on the free two-step group $N_{3,2}$. Its topological and homogeneous dimensions are respectively $d = 6$ and $Q = 9$. Write $\varsigma(L)$ for the infimum of all $s \geq 0$ for which a scale-invariant local W_2^s norm controls the weak- $(1, 1)$ norm of $F(L)$. The later paper also defines $\varsigma_-(L)$ using the formally weaker requirement of L^p boundedness for all $1 < p < \infty$, controlled by a local W_∞^s norm; standard interpolation gives $\varsigma_-(L) \leq \varsigma(L)$.

Source conjecture (Martini–Müller, arXiv:1210.8090, Theorem 1). *If*

$$\sup_{t>0} \|\eta F(t \cdot)\|_{W_2^s} < \infty \quad \text{for some } s > 3,$$

then $F(L)$ is of weak type $(1, 1)$ and bounded on every $L^p(N_{3,2})$, $1 < p < \infty$. The authors conjecture that the condition $s > d/2 = 3$ is sharp. Equivalently, their theorem gives $\varsigma(L) \leq 3$.

Answering theorem (Martini–Müller, arXiv:1508.01687, Theorem 3). *For every homogeneous sub-Laplacian $\mathcal{L}$ on a two-step stratified group of topological dimension d and homogeneous dimension Q ,*

$$\frac{d}{2} \leq \varsigma_-(\mathcal{L}) \leq \varsigma(\mathcal{L}) \leq \varsigma_+(\mathcal{L}) < \frac{Q}{2}.$$

Applying the answering theorem with $d = 6$ yields

$$3 \leq \varsigma_-(L) \leq \varsigma(L).$$

Together with the source upper bound, this proves the exact characterization

$$\boxed{\varsigma_-(L) = \varsigma(L) = 3.}$$

Thus no multiplier theorem of the indicated uniform type can hold below $s = 3$, even after dropping weak type $(1, 1)$ and replacing the W_2^s hypothesis by the stronger local W_∞^s hypothesis.

Scope caveat. The source’s broader question—whether topological dimension always replaces homogeneous dimension for arbitrary stratified groups—is not settled here. The answering paper covers all two-step groups and explicitly leaves the higher-step extension as an open problem requiring new ideas.

References

- [1] A. Martini and D. Müller, *L^p spectral multipliers on the free group $N_{3,2}$* , arXiv:1210.8090; Studia Math. 217 (2013), 41–55.
- [2] A. Martini and D. Müller, *Spectral multipliers on 2-step groups: topological versus homogeneous dimension*, arXiv:1508.01687; Geom. Funct. Anal. 26 (2016), 680–702.